Modernizing How Interval Estimation is Taught in Undergraduate Statistics Courses

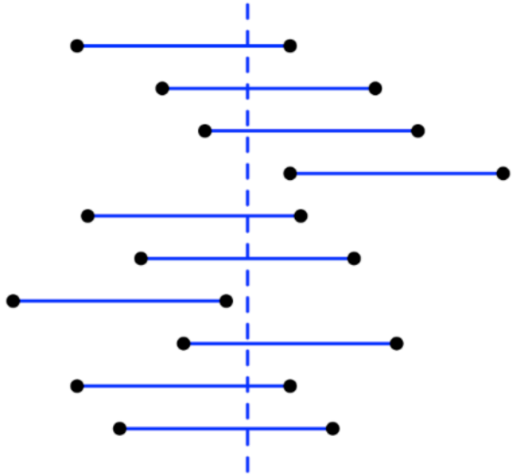
Peter E. Freeman - Department of Statistics & Data Science - Carnegie Mellon University

2025-08-07

What This Presentation is About (and What it is Not About)

- Upon reflection, we have decided that a more precise title for this presentation might be
  - **Modernizing How Students Compute Interval Estimates in Mathematical Statistics Courses**
- So, this presentation is about…
  - …introducing general numerical methods for interval estimation that supersede the use of, e.g., the pivotal method and statistical tables
  - …asking how we might utilize these methods in introductory settings
- …while it is not about…
  - …teaching students about what interval estimates represent and how they are to be interpreted
  - …quantitative assessment of student learning (so…treat this presentation as an editorial!)

Not this…



…but this…

```
f <- function(mu,y.obs,q,s2,n) {  # user-defined function to
  t.obs = (y.obs - mu)/sqrt(s2/n) # evaluate F_Y(y_obs|mu)-q=0
  pt(t.obs,n-1) - q
}
uniroot(f,interval=c(-100,100),y.obs=2.3,q=0.975,s2=1.1,n=15)$root
uniroot(f,interval=c(-100,100),y.obs=2.3,q=0.025,s2=1.1,n=15)$root
```

The Setting

- The numerical methods that we present in this presentation have been taught to students since Fall 2022 in the Carnegie Mellon University classes *Probability and Statistical Inference I* and *II*, classes which…
  - …are targeted to statistics majors (and not non-majors who simply are seeking a probability course)
  - …are calculus-based mathematical statistics courses with technically no statistics pre-requisites (although virtually all students had taken at least one algebra-based introductory class beforehand)
  - …utilize spiral learning - "statistics for probability" as opposed to "statistics, then probability" - so students are exposed to core concepts methods multiple times over two semesters
  - …have ≈ 175 students per iteration

Modern Probability and Statistical Inference

Illustrated with R

Peter E. Freeman (Department of Statistics & Data Science, Carnegie Mellon University)

https://mpsibook.github.io/

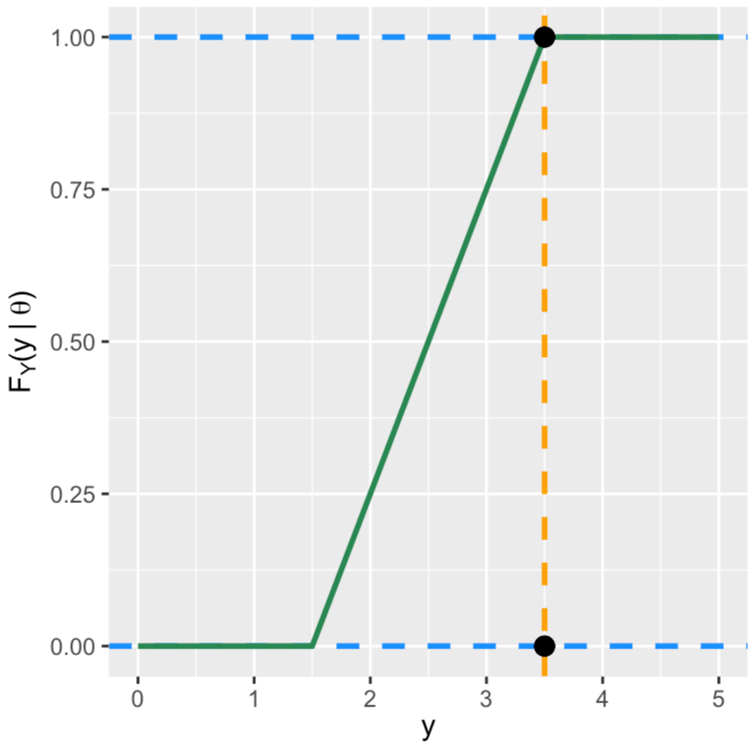
The Build-Up

- Prior to diving into interval estimation for the first time in *Probability and Statistical Inference I*, students are introduced to core concepts of probability and statistical inference, including *cumulative distribution functions (cdfs)*; *statistics*; and *sampling distributions*

- We introduce interval estimation by positing the following situation:
  - we collect independent and identically distributed (iid) data $\mathbf{X} = \{X_1, \ldots, X_n\}$
  - we compute a statistic $Y = g(\mathbf{X})$, with our observed value being y_{obs}
  - we determine that Y is sampled according to a $\text{Uniform}(\theta - 1, \theta + 1)$ distribution (this is admittedly unrealistic, but fine for intuition-building)
- We then ask: what is a *plausible* interval for θ, given y_{obs}?
- One way to answer this is to invert each of the following CDF-based equations, solving for θ:

$$F_Y(y_{\text{obs}}|\theta) = 0 \quad \text{and} \quad F_Y(y_{\text{obs}}|\theta) = 1$$

  - $F_Y(\cdot)$ is the cdf for the statistic Y; here, $F_Y(y|\theta) = (y - \theta + 1)/2$
  - solving the two equations above, we find that $\theta = y_{\text{obs}} + 1 \;\text{ and }\; y_{\text{obs}} - 1$, respectively
  - our "plausible interval" is thus $[y_{\text{obs}} - 1, y_{\text{obs}} + 1]$ - heck, this is easy, right?
  - (the figure at right shows the cdf [green line] for $\theta = 2.5$, the lower bound on θ when $y_{\text{obs}} = 3.5$…we can see that the line passes through the black dot at $(y_{\text{obs}} = 3.5, 1)$)



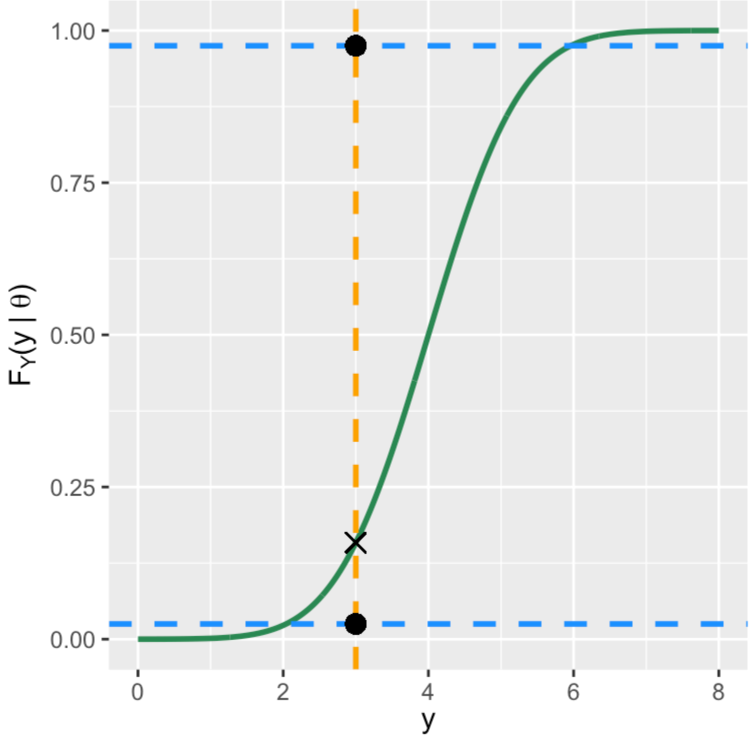
The Build-Up, Part II

- An issue with the algorithm above is that it only provides a “useful” interval when the sampling distribution is a finite, bounded distribution
- So what do we do in more general situations?
  - we first combine the two equations above into a single equation:

$$F_Y(y_{\rm obs}\,|\,\theta) - q = 0$$

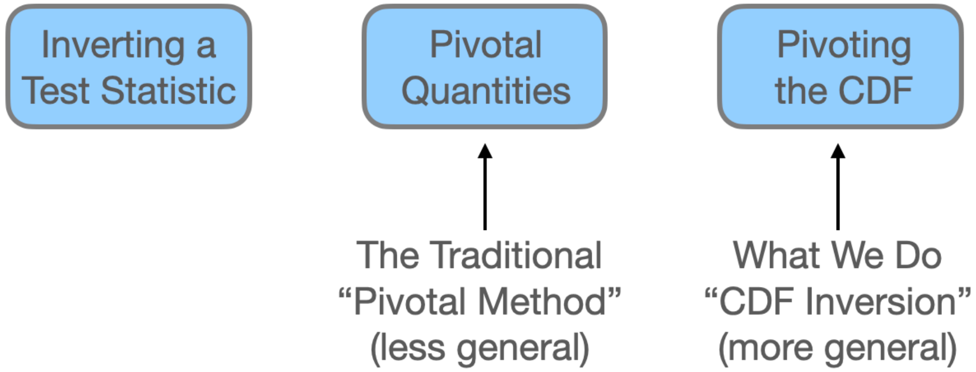
  - we then appeal to convention and set q to, e.g., $\alpha/2$ and $1 - \alpha/2$, where typically $\alpha = 0.05$
- We can now determine, e.g., 95% confidence intervals for θ, in any situation in which we know, or *can simulate* (more on this later!), the sampling distribution for Y

- Assume that $Y \sim N(\theta, 1)$ and that $y_{\rm obs} = 3$ and we wish to construct a 95% interval
  - as illustrated at right, we would change θ until the green curve (the cdf) passes through the black dot at $(y_{\rm obs} = 3, q = 0.025)$ (for the upper bound) and the one at $(y_{\rm obs} = 3, q = 0.975)$ (for the lower bound)
- Note: if we are unwilling or unable to specify the distribution according to which the individual X_i’s are sampled, we might apply the *central limit theorem* or shift to the separate but related concept of *bootstrapping* (more on this later)



“Wait…This Doesn’t Look At All Like What I’m Used To”

- That’s because it probably isn’t
- Casella & Berger (2002) present three separate algorithms for estimating intervals:



- The typical algorithm presented in mathematical statistics textbooks is the “pivotal method,” in which we define a quantity (a pivot) that is (a) a function of both y_{obs} and θ and (b) has a sampling distribution that does not depend on θ
  - primary issue: pivots generally do not exist for arbitrary distributions, making the pivotal method insufficiently general
- Our “CDF inversion” technique (so-called so as not to reuse “pivot”) has the benefit that “even if $[F_Y(y_{\text{obs}}|\theta) - q = 0]$ cannot be solved analytically, we really only need to solve [it] numerically since the proof that we have a $1 - \alpha$ confidence interval [does] not require an analytic solution”
  - primary benefit: if we can specify or estimate the sampling distribution of Y, we can apply the CDF inversion method (hence this method is more general)

Example 1: Numerical Solution

Let's suppose that we sample $n = 15$ iid data according to a normal distribution with unknown mean μ and unknown variance σ^2, and we adopt $Y = \bar{X}$ as our statistic. What is a 95% confidence interval for μ, given that we observe $y_{\rm obs} = \bar{x}_{\rm obs} = 2.3$ and $s^2_{\rm obs} = 1.1$?

- The relevant sampling distribution is

$$\frac{\bar{X} - \mu}{s_{\rm obs}/\sqrt{n}} = T \sim t_{n-1} \,,$$

  where t_{n-1} is a t-distribution for $n-1$ degrees of freedom

  - we cannot solve $F_Y(y_{\rm obs}|\mu) - q = 0$ by hand
  - because the equation has just one solution, we utilize R's `uniroot()` function

```
f <- function(mu,y.obs,q,s2,n) {  # user-defined function to
  t.obs = (y.obs - mu)/sqrt(s2/n) # evaluate F_Y(y_obs|mu)-q=0
  pt(t.obs,n-1) - q
}
uniroot(f,interval=c(-100,100),y.obs=2.3,q=0.975,s2=1.1,n=15)$root
```

```
[1] 1.719172
```

```
uniroot(f,interval=c(-100,100),y.obs=2.3,q=0.025,s2=1.1,n=15)$root
```

```
[1] 2.880836
```

- `interval`: this defines lower and upper bounds over which to search for the root
- The 95% confidence interval is $[1.719, 2.881]$
  - this is identical to $\bar{X} \pm t_{1-\alpha/2,n-1} \frac{s_{\rm obs}}{n}$ (as it should be!)

Example 2.0: Numerical Solution via Simulation

Let's suppose that we sample $n = 10$ iid data according to a beta distribution with unknown parameter a, where $b = 2.5$. We adopt $Y = \bar{X}$ as our statistic. (Note: this is *not* a sufficient statistic for a.) What is a 95% confidence interval for a, given that we observe $y_{\rm obs} = \bar{x}_{\rm obs} = 0.7$?

- The key here is that we do *not* know the sampling distribution for $\bar{X}$
  - however, because we are given that $X_i \sim {\rm Beta}(a, 2.5)$, we can estimate the sampling distribution for $\bar{X}$ on the fly!
  - (*important: we do not need to fall back upon bootstrapping!*)
- We can apply CDF inversion in all situations where we have R code for sampling data according to a distribution, or where we can code our own sampler

```
f <- function(a,y.obs,q,b,n,num.sim=10000,set.seed=101) {
  X <- matrix(rbeta(num.sim*n,a,b),nrow=num.sim)
  Y <- rowMeans(X)
  sum(Y<y.obs)/num.sim - q
}
uniroot(f,interval=c(0,100),y.obs=0.7,q=0.975,b=2.5,n=10)$root
```

```
[1] 3.549452
```

```
uniroot(f,interval=c(0,100),y.obs=0.7,q=0.025,b=2.5,n=10)$root
```

```
[1] 8.753047
```

Example 2.1: Numerical Solution via Simulation II (Order Statistics)

Let's suppose that we sample $n = 10$ iid data according to a beta distribution with unknown parameter a, where $b = 2.5$. We adopt $Y = \text{median}(X)$ as our statistic. (Note: this is also not a sufficient statistic for a.) What is a 95% confidence interval for a, given that we observe $y_{\text{obs}} = 0.7$?

- The only difference from the last example: we replace

```
Y <- rowMeans(X)
```

with

```
Y <- apply(X,1,median)
```

- As we (should!) expect, the interval given the sample median is wider than that given the sample mean
- *(And again…no bootstrapping!)*

```
f <- function(a,y.obs,q,b,n,num.sim=10000,set.seed=101) {
  X <- matrix(rbeta(num.sim*n,a,b),nrow=num.sim)
  Y <- apply(X,1,median)
  sum(Y<y.obs)/num.sim - q
}
uniroot(f,interval=c(0,100),y.obs=0.7,q=0.975,b=2.5,n=10)$root
```

```
[1] 2.917728
```

```
uniroot(f,interval=c(0,100),y.obs=0.7,q=0.025,b=2.5,n=10)$root
```

```
[1] 8.896765
```

Example 3: Numerical Solution for a Discrete Distribution

Let's suppose that we sample $n = 10$ iid data according to a Poisson distribution with unknown parameter λ. We adopt $Y = \sum_{i=1}^{n} X_i$ as our statistic. What is a 95% confidence interval for λ, given that we observe $y_{\rm obs} = \sum_{i=1}^{n} x_i = 66$?

- The sampling distribution for $Y = \sum_{i=1}^{n} X_i = n\bar{X}$ is $\text{Poisson}(n\lambda)$
  - after generating an interval estimate for $n\lambda$, we simply divide the endpoints by n
  - note that we have to apply a "discreteness correction" when we compute the lower bound, by inputting $y_{\rm obs} \rightarrow y_{\rm obs} - 1$, so that the underlying computation works out ("do we include the mass at $y_{\rm obs}$, or do we not?")

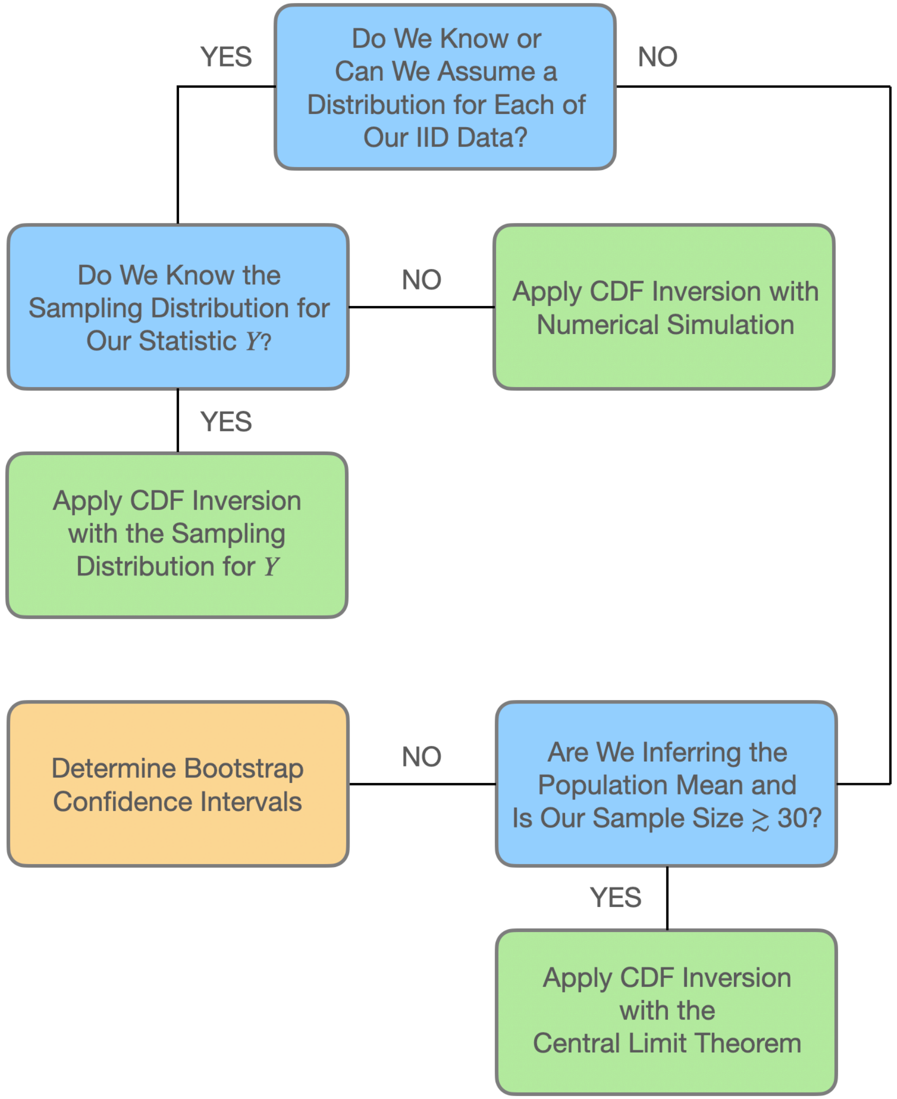
```r
f <- function(lambda,y.obs,q,n) {
  ppois(y.obs,n*lambda) - q
}
uniroot(f,interval=c(0,200),y.obs=66-1,q=0.975,n=10)$root / 10
```

```
[1] 0.5104439
```

```r
uniroot(f,interval=c(0,200),y.obs=66  ,q=0.025,n=10)$root / 10
```

```
[1] 0.8396832
```

The Interval Estimate Flowchart



So…What's the Upshot of All of This?

- CDF inversion is a general, easy-to-follow recipe for estimating intervals, either analytically (in relatively limited cases and not demonstrated here) or numerically (in all cases)
  - what we need: a statistic, its observed value, and either its sampling distribution or code that allows us to sample individual data so as to estimate the sampling distribution - full stop
  - for those for whom working with the mathematics is important: the computation of analytic solutions can touch upon, e.g., the method of moment-generating functions, the computation of cumulative distribution functions, working with order statistics, etc., etc. (note that one way we "cheat" so that we can broaden the pool of problems with analytic solutions is to construct intervals given one datum)
  - note that while we do not discuss one-sided intervals or intervals generated in situations where statistic values *decrease* (on average) as θ increases (e.g., when data are sampled according to the negative binomial distribution), nothing changes accept how we define the value of q in the equation $F_Y(y_{\rm obs}|\theta) - q = 0$

Interval Type	$E[Y]$ Increases With θ?	q for Lower Bound	q for Upper Bound
two-sided	yes	$1-\alpha/2$	$\alpha/2$
	no	$\alpha/2$	$1-\alpha/2$
one-sided lower	yes	$1-\alpha$	–
	no	α	–
one-sided upper	yes	–	α
	no	–	$1-\alpha$

Short Digression: Is There a Package for CDF Inversion?

- The short answer: yes (and it is called `cdfinv`)

Confidence Interval Estimation via CDF Inversion

Documentation for package ‘cdfinv’ version 0.1.0

- DESCRIPTION file.
- User guides, package vignettes and other documentation.

Help Pages

cdfinv	Computation of confidence intervals via CDF inversion
cdfinv.sim	Computation of confidence intervals via simulations and CDF inversion
ci_plot	Plotting of confidence intervals derived via CDF inversion
pnormvar	Cumulative distribution function for t distribution
ptmean	Cumulative distribution function for t distribution
qnormvar	Inverse cumulative distribution function for chi-square distribution
qtmean	Inverse cumulative distribution function for t distribution

- Feel free to send bug reports and RFEs to me

Moving Beyond the Mathematical Statistics Course: Questions to Ponder

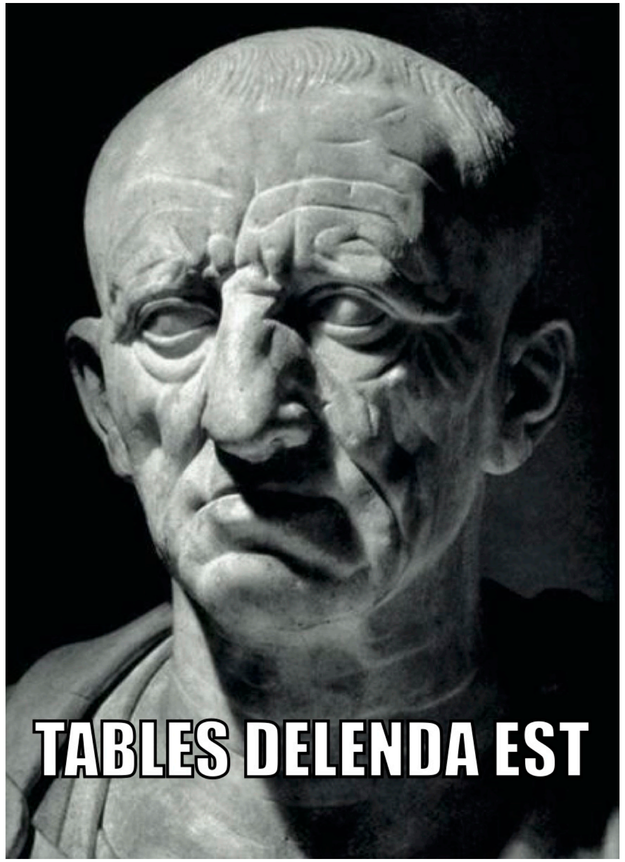
- In an introductory setting…
  - …is it better to provide students with a grab-bag of formulas like this, and expect by-hand computation?

$$\bar{X} \pm z_{1-\alpha} \frac{\sigma}{\sqrt{n}}$$

  - …or is it better to provide general-purpose apps built upon CDF inversion where the math is "abstracted away" and the job of the student is simply to map the information in word problems to the correct app settings?
  - (we are not claiming we know the answer)
- If we go with the apps: *how would they be best built so as to optimize student learning?*
- If we shift to the setting of an introductory data science course: to what extent should students code interval estimators from scratch…what is the cognitive cost relative to the cognitive benefit?
- More questions? Feel free to chime in!

However…There are Some Things (I Think) I Already Know…

- Statistical tables must be destroyed



- There is no such thing as “large-sample” and “small-sample” interval estimators…textbook writers: please stop saying that there is!

  Large-Sample Confidence Intervals 411
  Selecting the Sample Size 421
  Small-Sample Confidence Intervals for μ and $\mu_1 - \mu_2$ 425

- Related to this: students should not be taught how to construct intervals via the use of large-sample approximations!
  - they should construct “exact” intervals with software (and use the time not spent on a calculator reinforcing their knowledge of how to interpret the intervals they’ve just constructed)
- In the end: we should do things right, and not just do things “easy”
  - (although I’d argue that using grab-bag approximating formulas and statistical tables does not necessarily make computations any easier to carry out or interpret in the end)

Thank You

- This presentation is available at `https://github.com/pefreeman/Confidence-Intervals`

For That Person Who Will Ask About Python

- Here is, e.g., Example 1 done using Python

```
import numpy as np
from scipy import optimize
from scipy.stats import t

def f(mu, yObs, q, s2, n):
    return t.cdf((yObs-mu)/(np.sqrt(s2/n)), df = n-1) - q

s2     = 1.1
yObs   = 2.3
n      = 15

q = 0.975
l = optimize.root_scalar(f, bracket=[-1000,1000], args=(yObs, q, s2, n)).root
q = 0.025
u = optimize.root_scalar(f, bracket=[-1000,1000], args=(yObs, q, s2, n)).root
print(np.round([l,u],3))
```

```
[1.719 2.881]
```